

```
create database joinsdb;
use joinsdb;
```

```
CREATE TABLE department (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50),
location VARCHAR(50)
);
INSERT INTO department VALUES
(10,'HR','Jaipur'),
(20,'IT','Bangalore'),
(30,'Finance','Mumbai'),
(40,'Marketing','Delhi'),
(50,'Legal','Pune');
```

```
CREATE TABLE employee (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
dept_id INT,
salary INT,
city VARCHAR(50)
);
INSERT INTO employee VALUES
(101,'Amit',10,45000,'Jaipur'),
(102,'Bhavna',20,72000,'Bangalore'),
(103,'Chirag',20,68000,'Bangalore'),
(104,'Divya',30,55000,'Mumbai'),
(105,'Esha',NULL,39000,'Delhi'),
(106,'Farhan',40,61000,'Delhi'),
(107,'Gaurav',60,50000,'Kolkata'),
(108,'Hina',30,83000,'Mumbai');
```

```
CREATE TABLE projects (
proj_id INT PRIMARY KEY,
proj_name VARCHAR(50),
dept_id INT,
budget INT
);
INSERT INTO projects VALUES
(1,'Payroll Revamp',10,120000),
(2,'Cloud Migration',20,500000),
(3,'Mobile App',20,300000),
(4,'Audit Automation',30,250000),
(5,'Brand Refresh',40,180000),
(6,'Data Lake',70,400000);
```

```
select e.emp_id,e.emp_name,d.dept_name,d.location
from employee as e inner join department as d
on e.dept_id=d.dept_id;
```

```
select p.proj_id,p.proj_name,d.dept_name,p.budget
from department as d join projects as p
where d.dept_id=p.dept_id;
```

```
select e.emp_name,d.dept_name,p.proj_name
from employee as e join department as d join projects as p
on e.dept_id=d.dept_id and d.dept_id =p.dept_id;
```

```
select e.emp_id,e.emp_name,e.dept_id,d.dept_name
from employee as e left join department as d
on e.dept_id=d.dept_id;
```

```
select d.dept_id,d.dept_name,p.proj_name,p.budget
from department as d left join projects as p
on d.dept_id=p.dept_id;
```

```
select e.emp_id,e.emp_name,e.dept_id
from employee as e left join department as d
on e.dept_id=d.dept_id where d.dept_name is null;
```

```
select e.emp_name,d.dept_id,d.dept_name
from employee as e right join department as d
on e.dept_id=d.dept_id;
```

```
select d.dept_name,p.proj_id,p.proj_name,p.dept_id
from department as d right join projects as p
on d.dept_id=p.dept_id;
```

```
select * from employee as e cross join department as d
where (dept_name='Legal' or dept_name='Finance')
and salary>70000;
```

```
select p.proj_name,p.budget,d.dept_name,d.location
from projects as p cross join department as d
where budget>=400000 and location in('Mumbai','pune');
```

```
select count(*) from employee cross join department
on employee.dept_id=department.dept_id;
```

```
select e.emp_id, e.emp_name,e.dept_id,e.salary from employee as e
where salary >60000;
```

```
select e.emp_id,e.emp_name,e.salary,e.city
from employee as e
where city in('Delhi','Mumbai');
```

```
select e.emp_id, e.emp_name,e.dept_id,e.salary from employee as e
where dept_id is null;
```

```
select d.dept_id, d.dept_name, d.location
from department as d
where location in ('Banglore','Delhi');
```

```
select d.dept_id, d.dept_name, d.location
from department as d
where dept_id>=30;
```

```
select d.dept_id, d.dept_name, d.location
from department as d
where dept_name like (F%) or like(L%);
```

```
select p.proj_id,p.proj_name,p.dept_id,p.budget
from projects as p
where budget between 200000 and 400000;
```

```
select p.proj_id,p.proj_name,p.dept_id,p.budget
```

```
from projects as p  
where dept_id=20;
```

```
select p.proj_id,p.proj_name,p.budget  
from projects as p  
where (proj_name like('%a%') and budget<300000);
```